

Mohamed Badry

Machine Learning Engineer & Data Analyst

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🔗 Portofolio

CAREER DEVELOPMENT

AWS Machine Learning Engineer & Data Analyst with **1+ year of experience delivering actionable insights through advanced analytics, reporting, and ML solutions**. Skilled in **SQL, Python, and data visualization tools** to transform complex datasets into clear, data-driven decisions. Experienced in **end-to-end ML pipelines**, including data ingestion, feature engineering, model development, deployment, and monitoring using **MLOps best practices**. Passionate about optimizing processes and enabling **business impact through analytical insights and predictive modeling**.

WORK EXPERIENCE

AWS Machine Learning Engineer Trainee

11/2024 – 05/2024

National Telecommunication Institute (NTI)

Cairo, Egypt

- Designed, automated, and deployed **end-to-end ML pipelines on AWS SageMaker**, achieving **95%+ model accuracy**.
- Improved model performance by **20%** through **hyperparameter tuning, feature selection, and regularization**.
- Built **interactive Power BI dashboards** to track model performance, operational KPIs, and trends, leading to **enhanced monitoring and 15% faster anomaly detection**.

Lead, Data Science & Machine Learning Team

09/2023 – 04/2024

Cisco Networking Student Club

Cairo, Egypt

- Mentored **25+ students** on Python, EDA, and ML workflows, resulting in **production-ready analytical projects**.
- Led a team that improved model accuracy by **30% over baseline solutions** using **data-driven optimization strategies**.
- Organized and delivered **4+ AI training programs** on ML deployment, data visualization, and analytical best practices.

Machine Learning Intern

07/2023 – 07/2024

Instant Software Solutions

Cairo, Egypt

- Conducted **foundational data analysis** including **cleaning, feature engineering, and EDA** on datasets exceeding **100K+ records**, improving data quality and reliability.
- Built and optimized **supervised ML models** achieving **94% accuracy**.
- Automated data preprocessing pipelines, reducing manual effort by **40%** and improving **workflow scalability and reproducibility**.

SKILLS

- Analytics & Reporting : (SQL, Python, R, Excel, Power BI, Tableau, Matplotlib, Seaborn, Plotly)
- Machine Learning & AI : (Scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, PyTorch, AutoML, PyCaret)
- Cloud & Deployment(AWS (SageMaker, EC2, Lambda), Flask, FastAPI, MLflow)
- Databases (MySQL, SQL Server, Firebase)
- Core Concepts (Feature Engineering, Exploratory Data Analysis (EDA), Model Optimization, Cross-Validation, Statistical Inference)

PROJECTS

PathCoder - Career Path Recommender (Graduation Project) 🔗

09/2024 – 05/2025

- Developed an **AI-powered career guidance web app** predicting jobs and identifying key skills for developers using **job market data and 200K+ user profiles**.

- Trained a **Random Forest model** achieving **92% accuracy** with SHAP explainability to interpret recommendations.
- Built a **full-stack solution** using Flask, Streamlit, Firebase Auth, and MySQL.
- Implemented **MLflow** for workflow reproducibility (+40%) and added an **NLP chatbot & TensorFlow-based emotion detection** to enhance UX.

Technologies: Python, Scikit-learn, TensorFlow, Flask, Streamlit, MySQL, Firebase, MLflow

Predictive_Maintenance_NASA_turbofan

04/2025 – 05/2025

- Built a **predictive maintenance solution** to forecast engine failures and **Remaining Useful Life (RUL)** using NASA CMAPSS (FD001) sensor data.
- Engineered **critical features from time-series sensor data** and trained regression models, enabling proactive maintenance decisions.
- **Automated preprocessing and modeling pipelines** for efficiency and reproducibility.

Technologies: Time-series analysis, Feature Engineering, Regression Models

EDUCATION

BSc (Bachelor of Science) in Data Science

Arab Open University

09/2021 – 06/2025

cairo, egypt

CERTIFICATIONS

AWS Certified Machine Learning - Specialty

(practical experience building and deploying ML pipelines on AWS)

Supervised Machine Learning: Regression and Classification


Hands-on predictive modeling projects

Intermediate SQL Server

Practical database querying and data analysis experience

LANGUAGES

Arabic
Native

English
Advanced